

GEOS-MPM Examples Suite Report

Generated by examples/runExampleSuite

May 26, 2026

Running and rerunning

Each example directory contains a copyable `pfw_input*.py` and a `runProblem` wrapper. The wrapper stages an isolated case directory under `testRunDirectory`, copies the needed PFW files, runs the particle-file writer, submits GEOS, submits dependent post-processing, and copies figures/logs back to the example `output/` directory.

```
cd scripts/preProcessing/materialPointMethodParticleFileWriter/examples
./runExampleSuite all --python /usr/tce/bin/python3 --force
./runExampleSuite all --case borehole_Ghareb --python /usr/tce/bin/python3 --force
./runExampleSuite report --python /usr/tce/bin/python3
```

If output exists, `runProblem` prompts before overwriting. Use `--force` for noninteractive reruns. The suite records case-level failures, continues with remaining cases by default, and reports incomplete or interrupted Slurm jobs explicitly.

Timing and completion summary

Example	Status	GEOS state	GEOS wall	GEOS CPU	Post state	Post wall
brazilianDisk_FMPM	figures available					
brazilianDisk_BSpline	figures available					
brazilianDisk_FLIP	figures available					
brazilianDisk_PIC	figures available					
brazilianDisk_XPIC	figures available					
benchzy	figures available					
borehole_Ghareb	complete	COMPLETED	00:00:40	00:00:00	FAILED	00:00:00
collidingDisks	complete	COMPLETED	00:00:12	00:00:00	FAILED	00:00:01
elasticDisk	complete	COMPLETED	00:01:26	00:00:00	FAILED	00:00:00
copperFoamCompression2D	missing figures	COMPLETED	00:00:06	00:00:00	COMPLETED	00:00:07
copperFoamCompression2D	missing figures	COMPLETED	00:00:06	00:00:00	COMPLETED	00:00:07
pdc	complete	COMPLETED	00:09:55	00:00:00	FAILED	00:00:00

Incomplete, failed, or missing-output cases

copperFoamCompression – missing figures

GEOS job/state: 6076455 / COMPLETED

Post job/state: 6076456 / COMPLETED

Command: /usr/gapps/visit/bin/visit -nowin -cli -s /p/lustre1/homel1/geosxTests/copperFoamCompression2D/copp
Running: cli3.5.0 -nowin -s /p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoamCompression2D_vis
VisIt: Error - VisIt could not read from the file "/p/lustre1/homel1/geosxTests/copperFoamCompression2D/mpm_
The generated error message was:
There was an error opening /p/lustre1/homel1/geosxTests/copperFoamCompression2D/mpm_* database. It may be an

The following error(s) may be helpful in identifying the problem:

VisIt: Error - VisIt could not read from the file "/p/lustre1/homel1/geosxTests/copperFoamCompression2D/copp

The generated error message was:

There was an error opening /p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoamCompression2D_mpm.

VisIt tried using the following file format readers to open the file: Uintah, PDB, Silo

If you know the specific format reader VisIt should use to

for VisIt to try. This will possibly give more information on

the exact error.

Ignoring unknown VisIt/script arguments: --initial-variable velocityMagnitude --final-variable velocityMagni

Output directory: /p/lustre1/homel1/geosxTests/copperFoamCompression2D/visit_frames

Opening database: /p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoamCompression2D_mpm.visit

OpenDatabase returned failure for /p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoamCompression

Traceback (most recent call last):

File "/p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoamCompression2D_visit_driver.py", line

File "/p/lustre1/homel1/geosxTests/copperFoamCompression2D/visitRender_copperFoamCompression2D.py", line 6

File "/p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoam_visit_common.py", line 607, in main

File "/p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoam_visit_common.py", line 171, in open_

raise FileNotFoundError("No GEOS-MPM plot database found. Expected <run-dir>/siloFiles/mpm_cpdi_*")

FileNotFoundError: No GEOS-MPM plot database found. Expected <run-dir>/siloFiles/mpm_cpdi_*

copperFoamCompression2D – missing figures

GEOS job/state: 6076455 / COMPLETED

Post job/state: 6076456 / COMPLETED

Command: /usr/gapps/visit/bin/visit -nowin -cli -s /p/lustre1/homel1/geosxTests/copperFoamCompression2D/copp

Running: cli3.5.0 -nowin -s /p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoamCompression2D_vis

VisIt: Error - VisIt could not read from the file "/p/lustre1/homel1/geosxTests/copperFoamCompression2D/mpm_

The generated error message was:

There was an error opening /p/lustre1/homel1/geosxTests/copperFoamCompression2D/mpm_* database. It may be an

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VisIt tried using the following file format readers to open the file: Uintah, PDB, Silo

If you know the specific format reader VisIt should use to

for VisIt to try. This will possibly give more information on

the exact error.

Ignoring unknown VisIt/script arguments: --initial-variable velocityMagnitude --final-variable velocityMagni

Output directory: /p/lustre1/homel1/geosxTests/copperFoamCompression2D/visit_frames

Opening database: /p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoamCompression2D_mpm.visit

OpenDatabase returned failure for /p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoamCompression

Traceback (most recent call last):

File "/p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoamCompression2D_visit_driver.py", line

File "/p/lustre1/homel1/geosxTests/copperFoamCompression2D/visitRender_copperFoamCompression2D.py", line 6

File "/p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoam_visit_common.py", line 607, in main

File "/p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoam_visit_common.py", line 171, in open_

raise FileNotFoundError("No GEOS-MPM plot database found. Expected <run-dir>/siloFiles/mpm_cpdi_*")

FileNotFoundError: No GEOS-MPM plot database found. Expected <run-dir>/siloFiles/mpm_cpdi_*

brazilianDisk_FMPM

Plane-strain Brazilian disk compression with quartz damage, Weibull strength scaling, and CPDI particles. **Status:** figures available

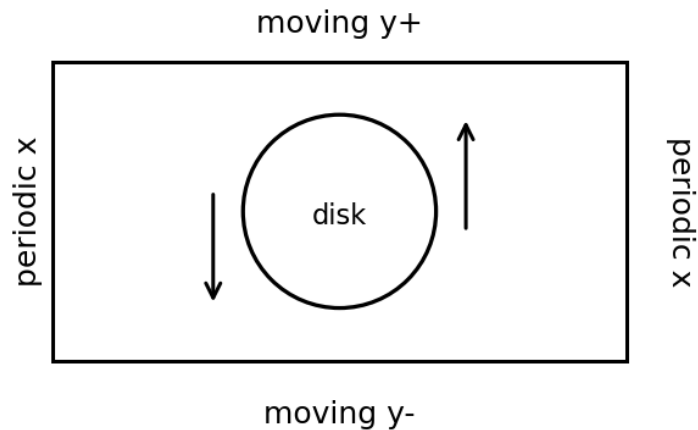


Figure 1: Schematic for brazilianDisk_FMPM.

Code features exercised.

- FMPM update
- CPDI particles
- quartz damage material
- moving y boundaries
- periodic x boundaries
- reaction history

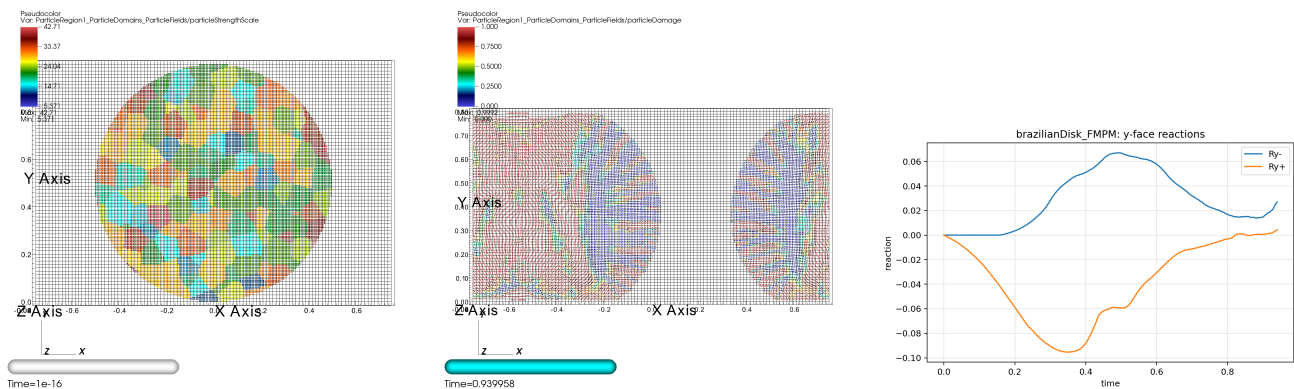


Figure 2: Initial state, final state, and reaction history for brazilianDisk_FMPM.

Output figures: brazilianDisk_FMPM_final_particleDamage.png, brazilianDisk_FMPM_initial_particleStrengthScale.png, brazilianDisk_FMPM_reactions_y.png

brazilianDisk_BSpline

Brazilian disk compression using the SinglePointBSpline particle type. The case keeps the FPM transfer path and quartz damage material while exercising the new particle interpolation option. **Status:** figures available

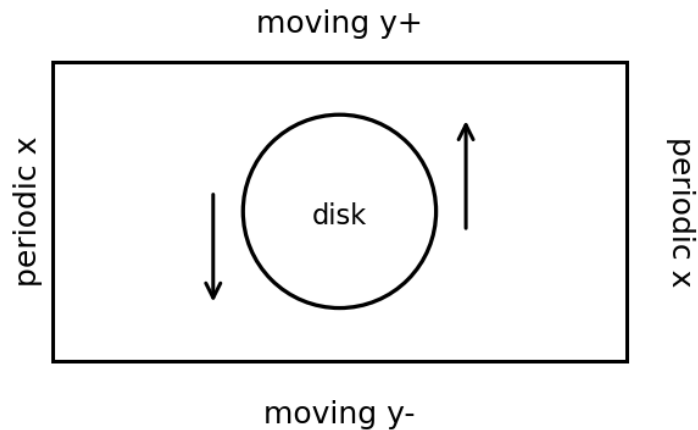


Figure 3: Schematic for brazilianDisk_BSpline.

Code features exercised.

- SinglePointBSpline particle type
- FPM update
- quartz damage material
- moving y boundaries
- periodic x boundaries
- reaction history

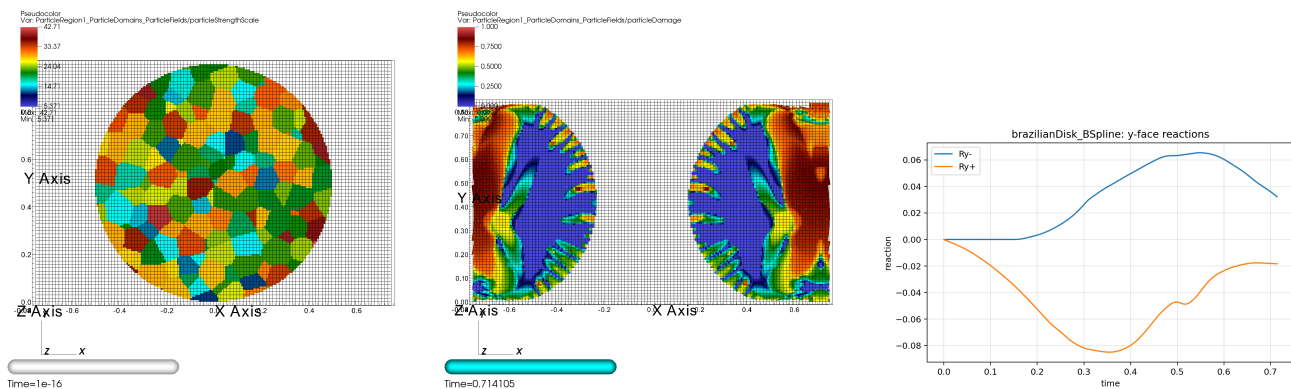


Figure 4: Initial state, final state, and reaction history for brazilianDisk_BSpline.

Output figures: brazilianDisk_BSpline_final_particleDamage.png, brazilianDisk_BSpline_initial_particleStrengthScale.png, brazilianDisk_BSpline_reactions_y.png

brazilianDisk_FLIP

Brazilian disk variant using FLIP grid-to-particle velocity transfer. **Status:** figures available

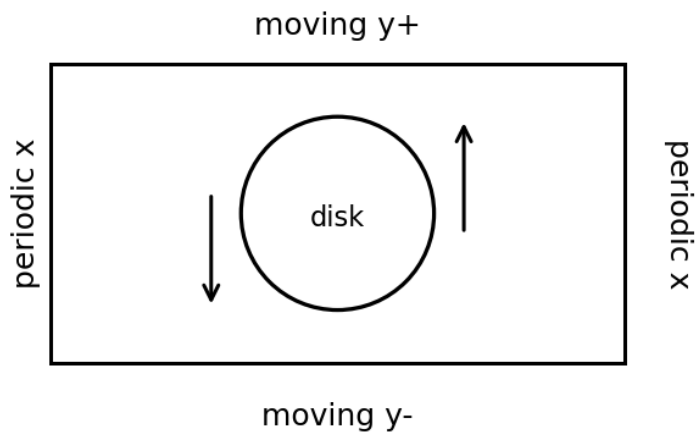


Figure 5: Schematic for brazilianDisk_FLIP.

Code features exercised.

- FLIP update
- quartz damage material
- moving y boundaries
- periodic x boundaries
- reaction history

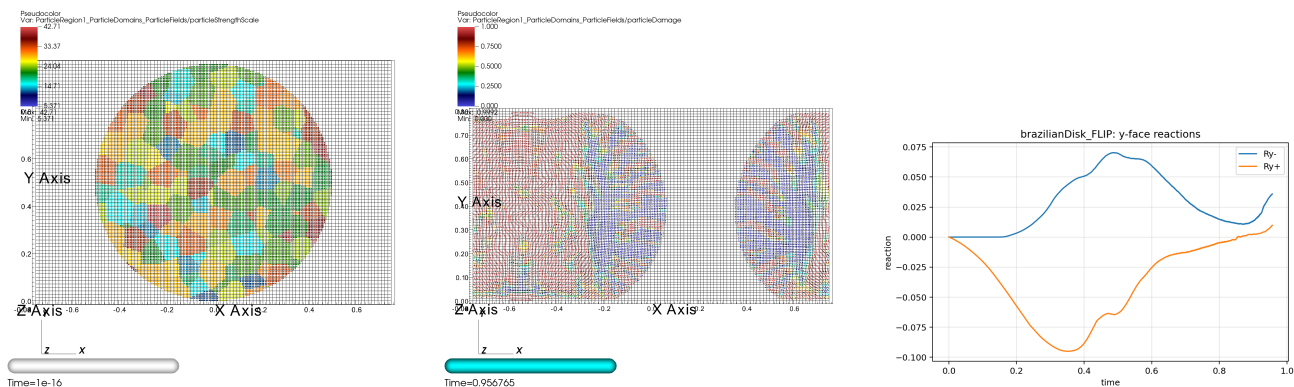


Figure 6: Initial state, final state, and reaction history for brazilianDisk_FLIP.

Output figures: brazilianDisk_FLIP_final_particleDamage.png, brazilianDisk_FLIP_initial_particleStrengthScale.png, brazilianDisk_FLIP_reactions_y.png

brazilianDisk_PIC

Brazilian disk variant using PIC velocity transfer. **Status:** figures available

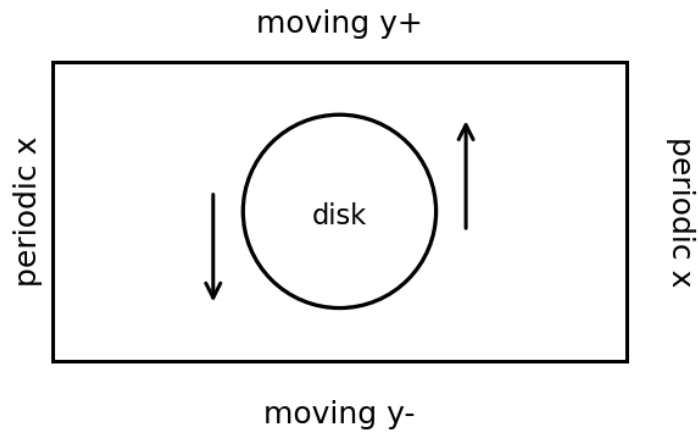


Figure 7: Schematic for brazilianDisk_PIC.

Code features exercised.

- PIC update
- quartz damage material
- moving y boundaries
- periodic x boundaries
- reaction history

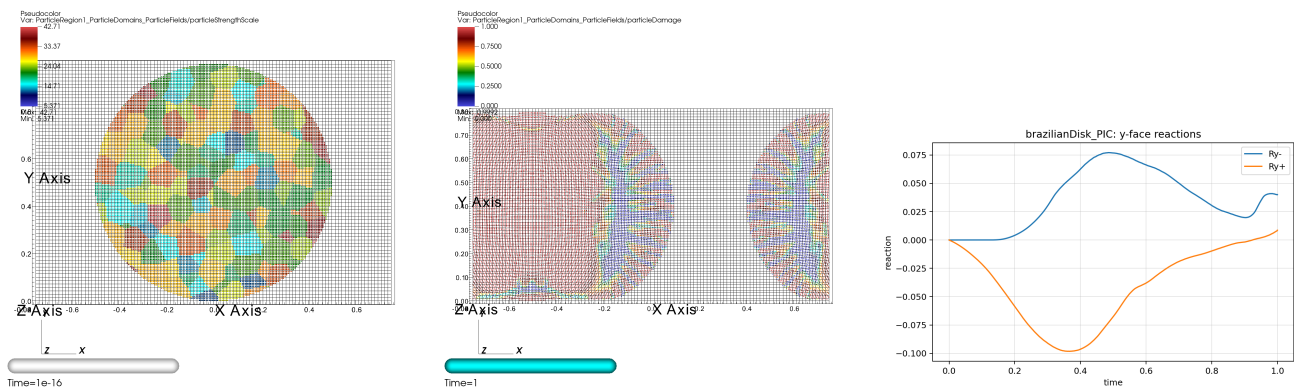


Figure 8: Initial state, final state, and reaction history for brazilianDisk_PIC.

Output figures: brazilianDisk_PIC_final.particleDamage.png, brazilianDisk_PIC_initial.particleStrengthScale.png, brazilianDisk_PIC_reactions.y.png

brazilianDisk_XPIC

Brazilian disk variant using XPIC velocity transfer. **Status:** figures available

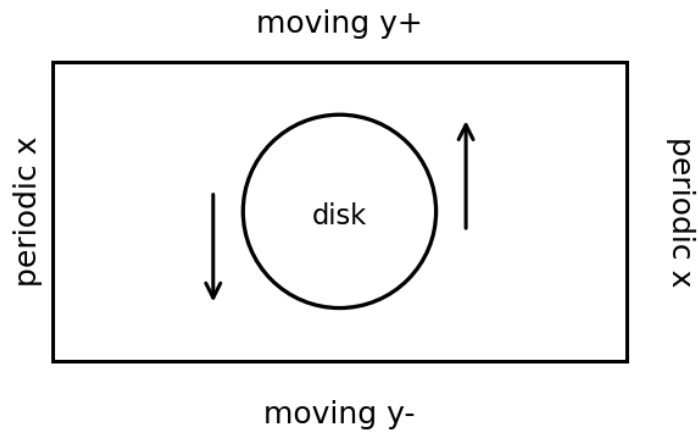


Figure 9: Schematic for brazilianDisk_XPIC.

Code features exercised.

- XPIC update
- quartz damage material
- moving y boundaries
- periodic x boundaries
- reaction history

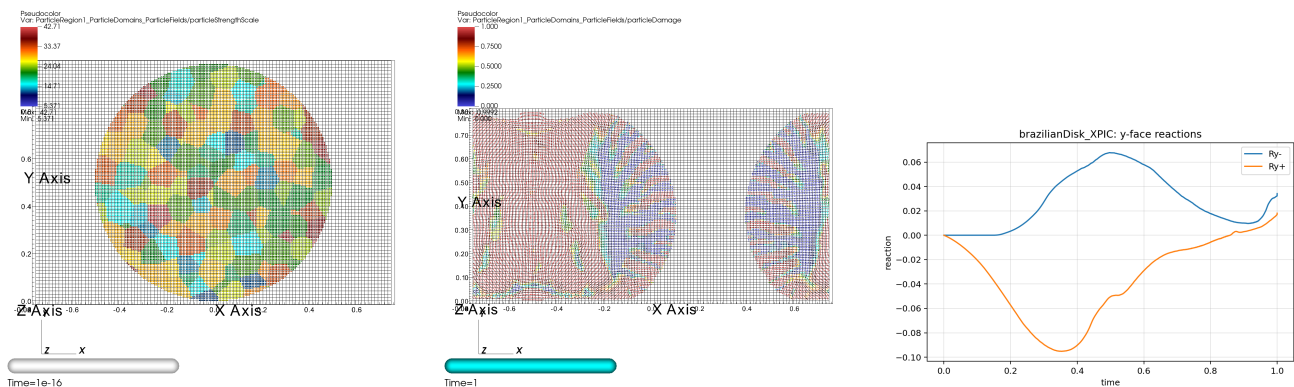


Figure 10: Initial state, final state, and reaction history for brazilianDisk_XPIC.

Output figures: brazilianDisk_XPIC_final_particleDamage.png, brazilianDisk_XPIC_initial_particleStrengthScale.png, brazilianDisk_XPIC_reactions_y.png

benchy

STL import and prescribed multi-field contact between a copper 3DBenchy boat and a steel ball. **Status:** figures available

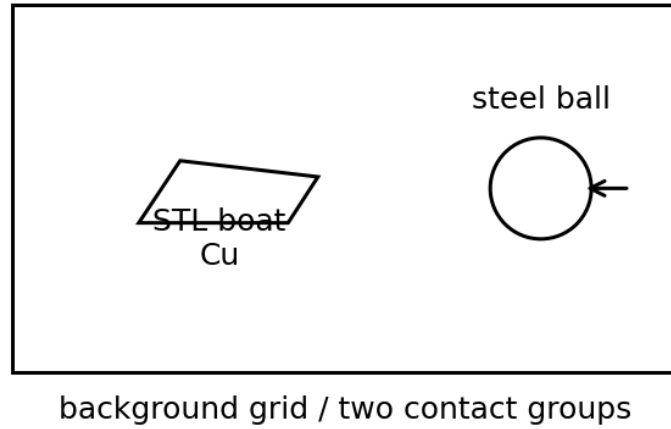


Figure 11: Schematic for benchy.

Code features exercised.

- STL geometry import
- copper boat and steel ball
- separate particle regions/contact groups
- prescribed multi-field contact
- multi-region VisIt rendering

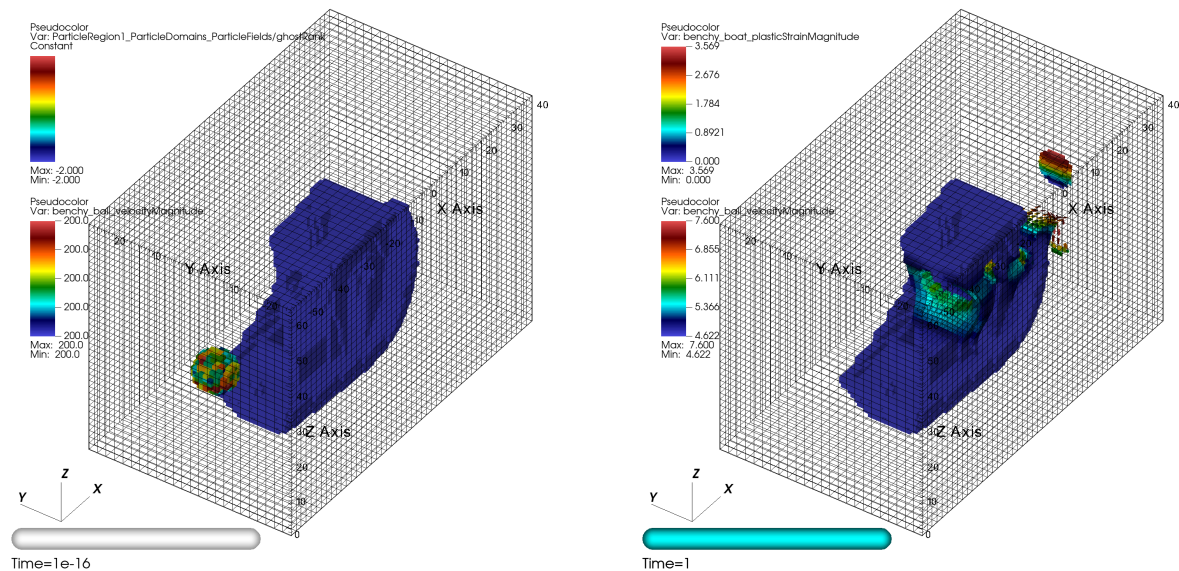


Figure 12: Initial and final state for benchy.

Output figures: benchy_FMPM_final_boatPlasticStrainMagnitude_ballVelocityMagnitude.png, benchy_FMPM_initial_boatGhost

borehole_Ghareb

A plane-strain borehole collapse demonstration using the Ghareb geomechanics parameterization from `pfw_materials.py`. The rendered initial state shows hydrostatic/mean pressure and the final state shows plastic-strain magnitude. **Status:** complete

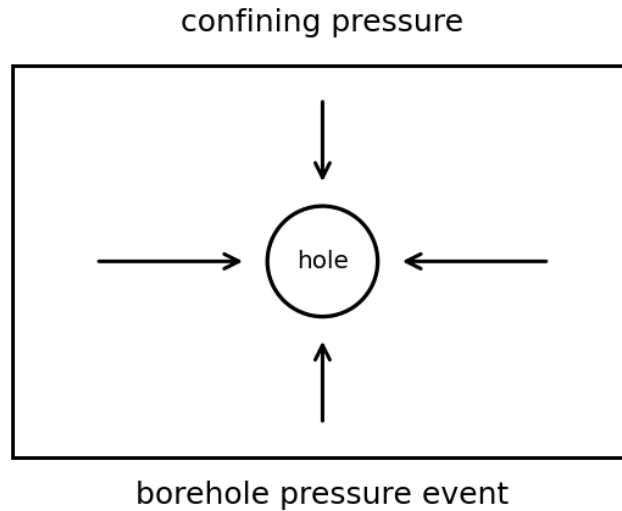


Figure 13: Schematic for borehole_Ghareb.

Code features exercised.

- Geomechanics material model
- InitializeStress MPM event
- confining-pressure event
- borehole-pressure event
- Voronoi-Weibull flaw field
- plane-strain CPDI

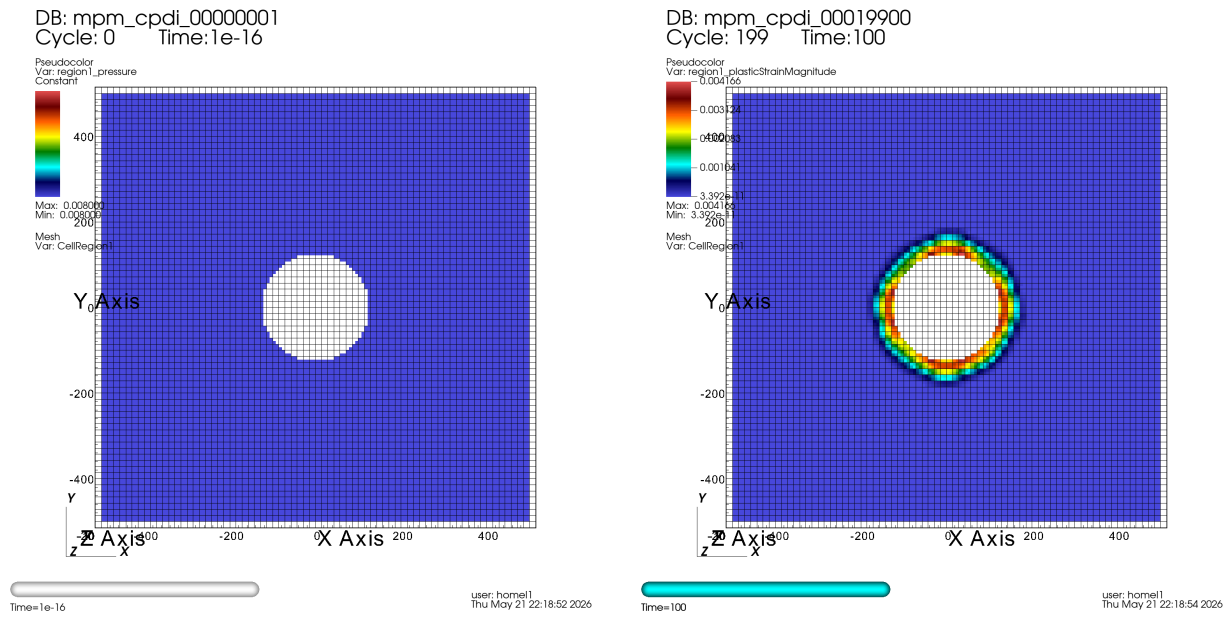


Figure 14: Initial and final state for borehole_Ghareb.

Output figures: borehole_Ghareb_final_plasticStrainMagnitude.png, borehole_Ghareb_initial_pressure.png

collidingDisks

Two disk-shaped particle bodies move toward one another in plane strain. The case exercises moving material points, contact-like interactions, and the XPIC update path. **Status:** complete

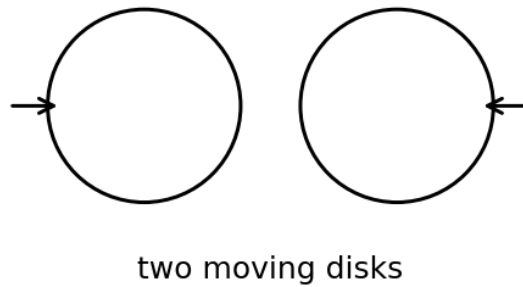


Figure 15: Schematic for collidingDisks.

Code features exercised.

- cylinder disk geometry
- XPIC update
- multi-object particle initialization
- box-average or reaction-style output when enabled

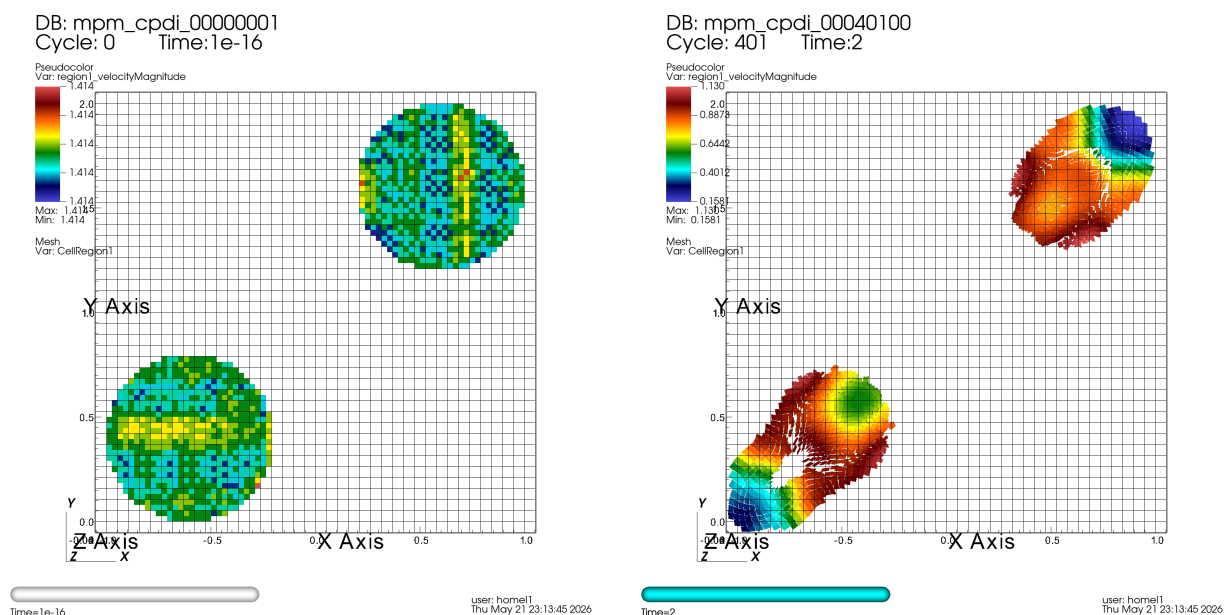


Figure 16: Initial and final state for collidingDisks.

Output figures: collidingDisks_final_velocityMagnitude.png, collidingDisks_initial_velocityMagnitude.png, collidingDisks_reaction

elasticDisk

An elastic disk compressed by prescribed boundary motion. It is a compact example of F-table interpolation, reaction history, and elastic response. **Status:** complete

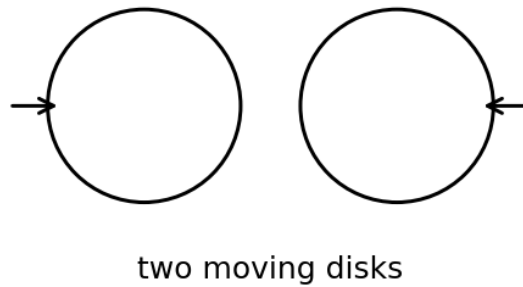


Figure 17: Schematic for elasticDisk.

Code features exercised.

- ElasticIsotropic material
- cosine F-table boundary motion
- reaction-history output when enabled
- plane-strain CPDI

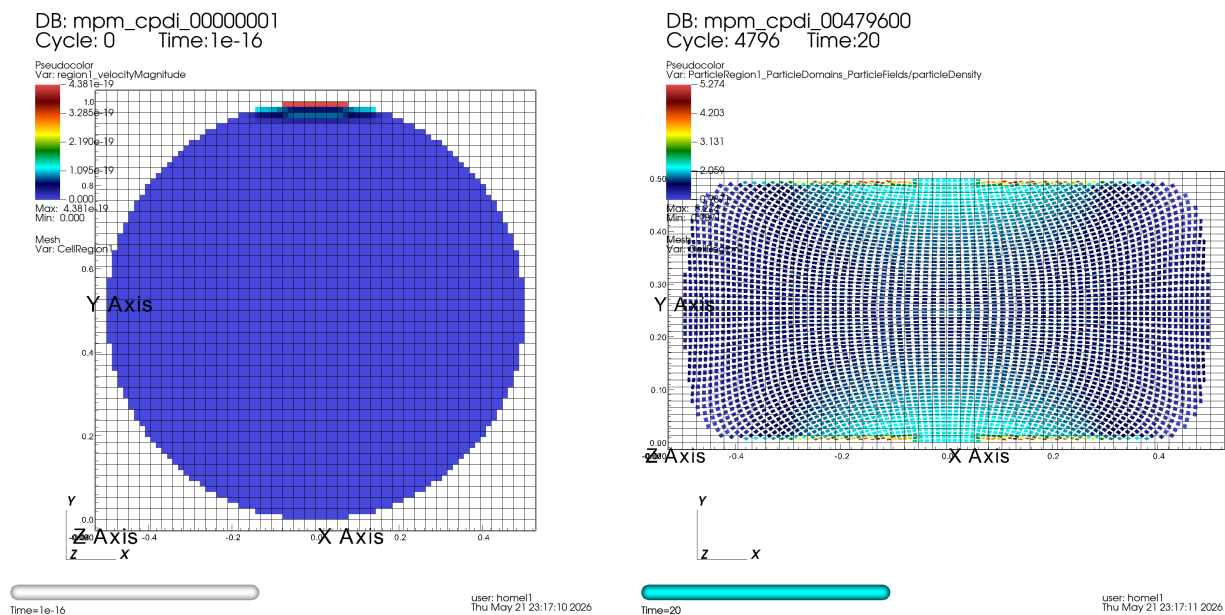


Figure 18: Initial and final state for elasticDisk.

Output figures: elasticDisk_final_density.png, elasticDisk_initial_velocityMagnitude.png, elasticDisk_reactions.png

copperFoamCompression

A 3D copper foam bar with periodic lateral pore structure impacts a symmetry plane in reverse-ballistic uniaxial compression. The initial and final visualizations show velocity magnitude. **Status:** missing figures

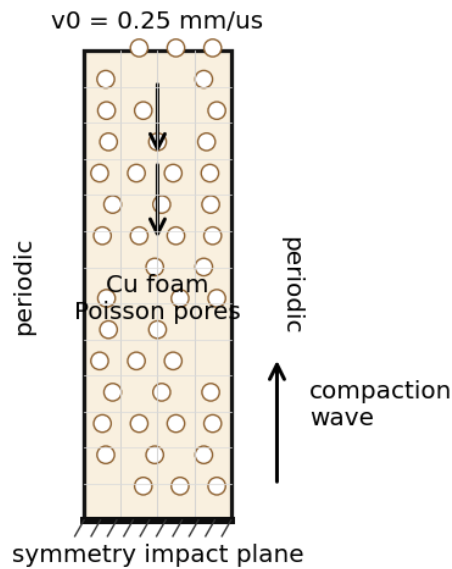


Figure 19: Schematic for copperFoamCompression.

Code features exercised.

- copper VonMisesJ material
- Poisson-disk spherical pores
- periodic x/y boundaries
- SinglePointBSpline particles
- compaction-wave stop-time estimate
- custom velocity-magnitude VisIt rendering

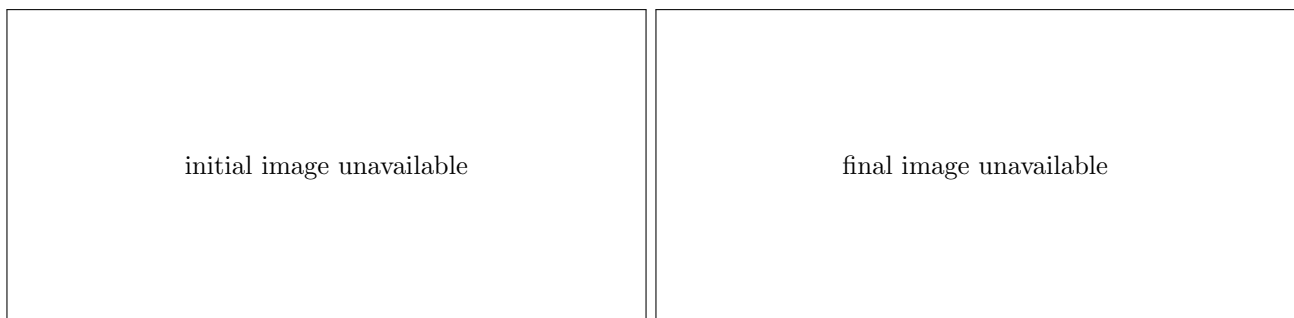


Figure 20: Initial and final state for copperFoamCompression.

Missing expected products: initial image, final image.

the exact error.

Ignoring unknown VisIt/script arguments: --initial-variable velocityMagnitude --final-variable velocityMagni

Output directory: /p/lustre1/home11/geosxTests/copperFoamCompression2D/visit_frames


```
Opening database: /p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoamCompression2D_mpm.visit
OpenDatabase returned failure for /p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoamCompression
Traceback (most recent call last):
  File "/p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoamCompression2D_visit_driver.py", line
  File "/p/lustre1/homel1/geosxTests/copperFoamCompression2D/visitRender_copperFoamCompression2D.py", line 6
  File "/p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoam_visit_common.py", line 607, in main
  File "/p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoam_visit_common.py", line 171, in open_
    raise FileNotFoundError("No GEOS-MPM plot database found. Expected <run-dir>/siloFiles/mpm_cpdi_*")
FileNotFoundError: No GEOS-MPM plot database found. Expected <run-dir>/siloFiles/mpm_cpdi_*
```

copperFoamCompression2D

A plane-strain copper foam bar with periodic x pore structure impacts a $y=0$ symmetry plane. The initial and final visualizations show velocity magnitude. **Status:** missing figures

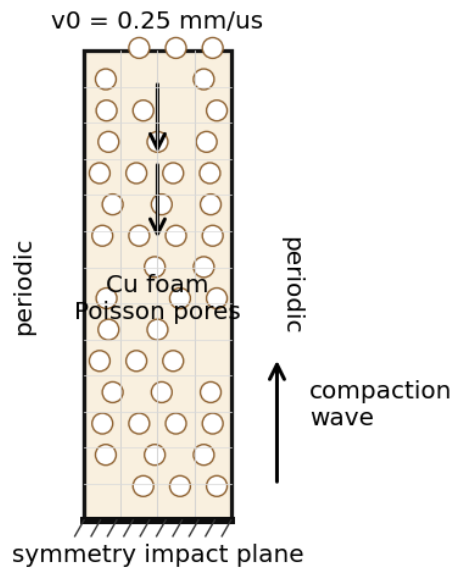


Figure 21: Schematic for copperFoamCompression2D.

Code features exercised.

- plane-strain MPM
- copper VonMisesJ material
- Poisson-disk circular pores
- periodic x boundary
- SinglePointBSpline particles
- compaction-wave stop-time estimate

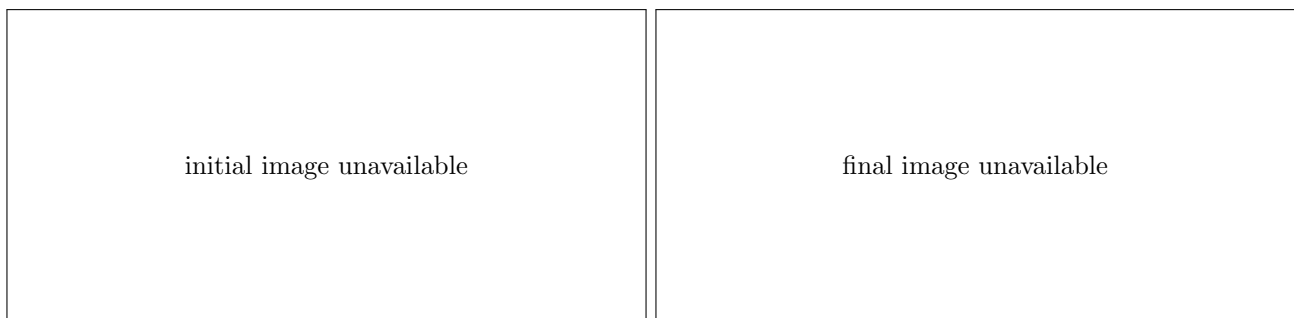


Figure 22: Initial and final state for copperFoamCompression2D.

Missing expected products: initial image, final image.

the exact error.

Ignoring unknown VisIt/script arguments: --initial-variable velocityMagnitude --final-variable velocityMagni

Output directory: /p/lustre1/home11/geosxTests/copperFoamCompression2D/visit_frames

```
Opening database: /p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoamCompression2D_mpm.visit
OpenDatabase returned failure for /p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoamCompression
Traceback (most recent call last):
  File "/p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoamCompression2D_visit_driver.py", line
  File "/p/lustre1/homel1/geosxTests/copperFoamCompression2D/visitRender_copperFoamCompression2D.py", line 6
  File "/p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoam_visit_common.py", line 607, in main
  File "/p/lustre1/homel1/geosxTests/copperFoamCompression2D/copperFoam_visit_common.py", line 171, in open_
    raise FileNotFoundError("No GEOS-MPM plot database found. Expected <run-dir>/siloFiles/mpm_cpdi_*")
FileNotFoundError: No GEOS-MPM plot database found. Expected <run-dir>/siloFiles/mpm_cpdi_*
```

pdc

A small PDC cutter example with a cutter interacting with a substrate. The final visualization uses the stored three-dimensional VisIt GUI view and overlays the background grid with plastic-strain-magnitude pseudocolors for both particle regions. **Status:** complete

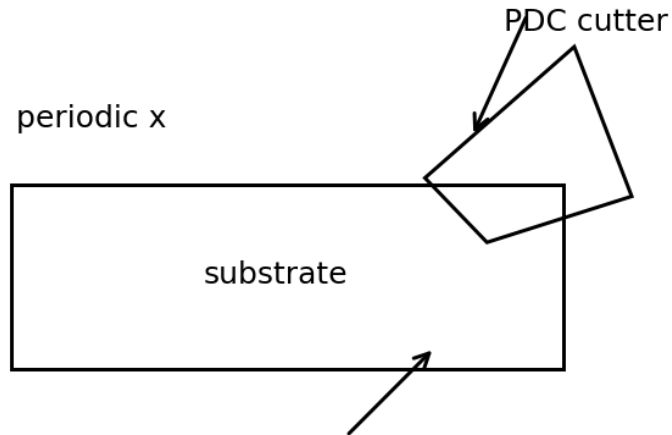


Figure 23: Schematic for pdc.

Code features exercised.

- periodic x boundary
- prescribed transverse boundary velocity
- PDC cutter/substrate geometry
- implicit contact gap correction
- two-particle-region plastic strain visualization
- stored 3D VisIt view

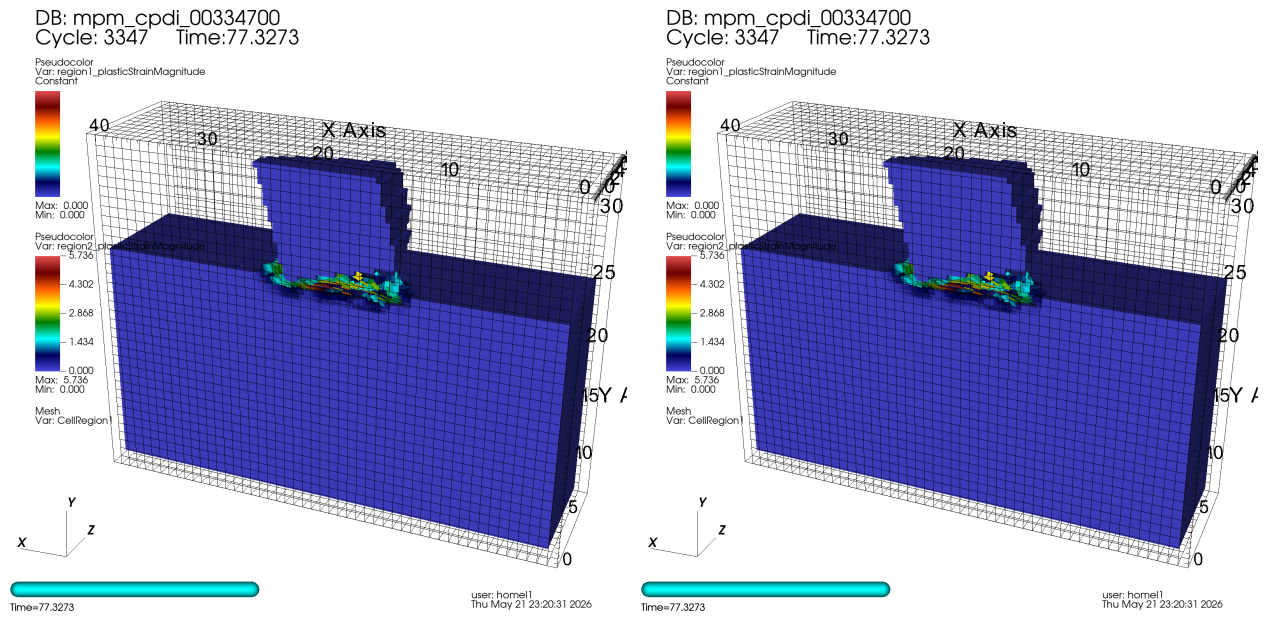


Figure 24: Initial and final state for pdc.

Output figures: pdc_final_plasticStrainMagnitude.png, pdc_initial_velocity.png